S2.1 The ionic model

Structure of ionic compounds

Learning outcomes

By the end of this section you should be able to:

- Describe the three-dimensional structure of an ionic compound.
- Predict the relative strength and an ionic bond by considering lattice enthalpy, ionic radius and charge.

Ionic minerals make up some of the most beautiful naturally occurring solids. Calcium carbonate is an ionic compound that comprises the main component in the shells of marine organisms, eggshells, rocks, pearls, stalagmites and stalactites. What is it about the structural arrangement of ions in the solid that allows for the formation of unique macroscopic structures?

Interactive 1 shows some examples of where calcium carbonate can be found in nature. Scroll through to see some naturally occurring calcium carbonate structures.

Interactive 1. Different Examples of Calcium Carbonate Found in Nature.

More information for interactive 1

An interactive slideshow with five slides presents photographs of naturally occurring calcium carbonate.
Users can switch between slides using arrows on the right and/or left. At the bottom of the page, there are five small circular indicators representing the total number of images in the interactive slideshow. Based on the image viewed, the specific indicator turns blue. The zoom-out arrow at the top right allows the users to view the images in full-screen mode.
Read below to know the various naturally occurring structures with calcium carbonate:
Slide 1: A group of mussel shells.
Slide 2: Several broken eggshells are placed on a brown wooden surface.
Slide 3: A rocky mountain next to a water body. The surface of the land and most of the mountain appears white.
Slide 4: A pearl inside an oyster shell.
Slide 5: Stalagmite and stalactite formations inside a cave.

In section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) you learned how the ionic bond is formed by the electrostatic attraction between oppositely charged ions. Does this attraction only happen between two oppositely charged ions or can it happen between many ions of opposite charge? This electrostatic attraction occurs between many ions in three-dimensions in a predictable arrangement to create a highly ordered solid called a lattice. All ionic compounds exist as three-dimensional lattice structures.

Interactive 2 shows how the lattice structure of lithium fluoride is formed from its elements. Press the 'next' button to watch the formation of an ionic lattice of LiF from its elements.

Interactive 2. Formation of an ionic lattice of LiF from its elements.

More information for interactive 2

An interactive animated slideshow displays the formation of a lattice structure of lithium fluoride. Next and previous buttons are on the bottom left allowing navigation between slides.
Slide 1: Fluorine is a gas and has atoms that exist in pairs. A balloon depicts pairs of fluorine atoms. Lithium is a solid metal arranged with lithium atoms.
Slide 2: One lithium atom can transfer its valence electron to one fluorine atom. Two units of Li^+ and F^- are formed after the electron transfer. Li^+ ions are smaller than F^- ions. Each F^- ion is attracted to one F^- ion. Ionic bonds are formed between oppositely charged ions.
Slide 3. The ions in the pairs experience an electrostatic attraction to all ions of opposite charge. Two units of Li^+ ion and F^- ion interact with each other.
Slide 4. All the formed ions self-assemble into a crystal lattice structure with the oppositely charged ions. The ions are arranged alternatively in the lattice.
Slide 5. This process happens with all available ions in three dimensions to form a cubic

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lattice.
A play/pause button is located at the bottom left to control the animation. An option for fullscreen is available at the bottom right.
This interactivity helps users understand the formation of LiF through ionic bonding and the electrostatic attraction between lithium and fluoride ions.

In **Interactive 2**, you can see that the ionic lattice of lithium fluoride is a three-dimensional repeating arrangement of ions, forming a crystalline solid. Notice how the formula of lithium fluoride is LiF, even though the structure of the compound contains many ions. It is important to remember that the chemical formula for ionic compounds is an empirical formula, which represents the simplest ratio of ions that make up that compound. In a three-dimensional structure of LiF, there are an equal number of Li^+ ions as there are F^- ions.

Let's look at two different lattice structures below in **Interactive 3**. Rotate and zoom the graphics to see the lattice structures for halite (NaCl) and fluorite (CaF_2) in more detail.

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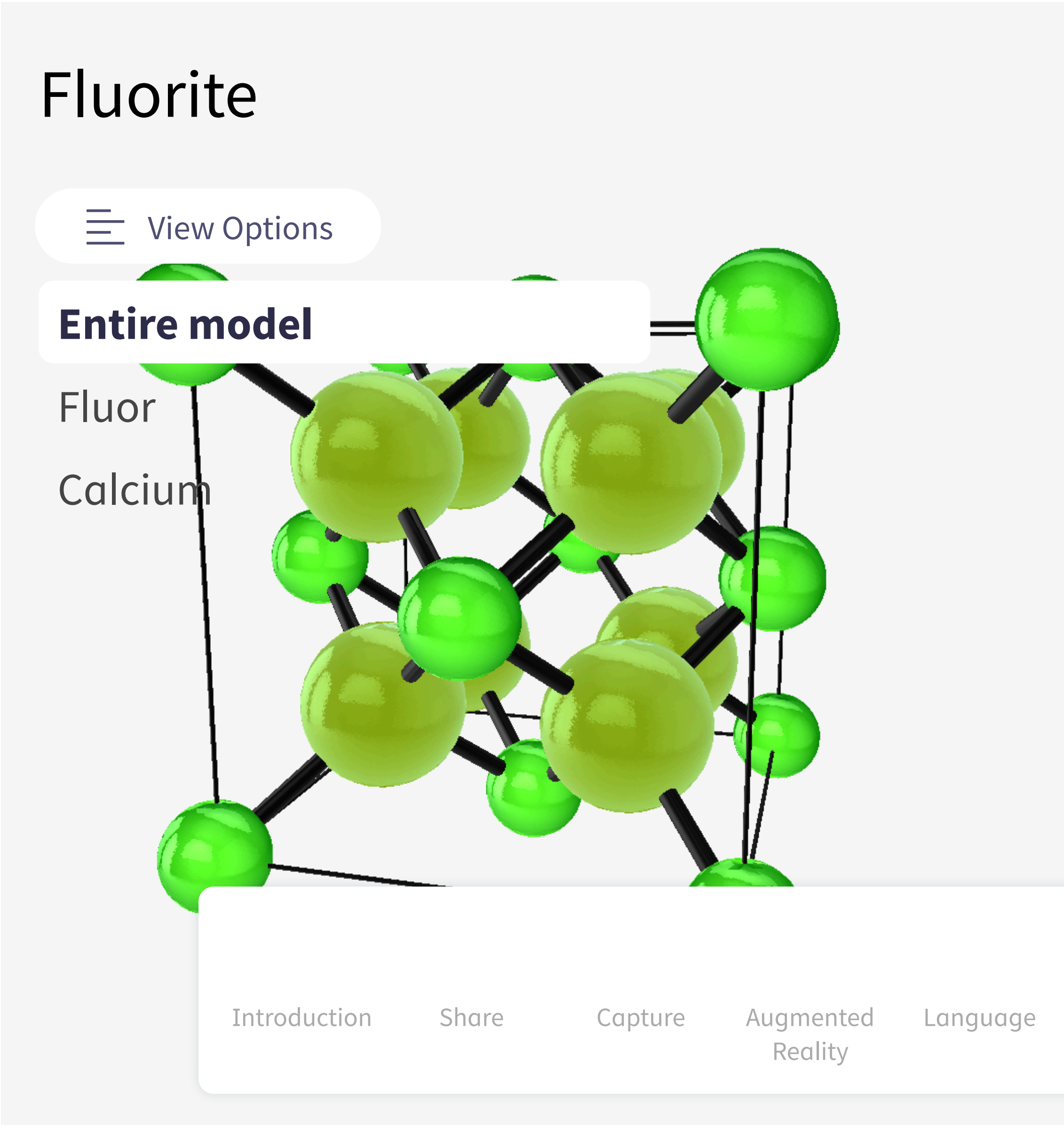
Interactive 3. Structure of Halite ($NaCl$).

More information for interactive 3

An interactive three-dimensional model depicts a ball-and-stick model of halite ($NaCl$). This model features a cubic structure where sodium ions occupy the corners and the centers of each face of the cube and chlorine ions occupy the center of each edge and the center of the cube.
Users can rotate the model to view it from different angles and zoom in or out using the mouse scroll. In the model, the sodium and chlorine ions are displayed as smaller and larger spheres, respectively.
A panel on the left provides view options, including the entire model, chlorine, and sodium. Selecting chlorine or sodium highlights their arrangement within the model.
On the right side, option for changing the background color and option for zooming the interactive are given.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction option displays the following details:
Chemical name: Sodium chloride
General formula: $NaCl$
Molar mass: 58.44 g/mol

Mohs scale hardness: 2
Crystal system: Cubic
Unit cell: Face-centered cubic lattice, with Na^+ and Cl^- in alternating positions.
Crystal habit: Predominantly forms cubic crystals.
Description: Pure sodium chloride, also known as halite in its natural form, is a colorless crystalline solid with a salty taste. It readily dissolves in water to produce brine, an electrically conductive solution.
Applications: Sodium chloride is used in the food industry as a seasoning and preservative and in the chemical industry as a raw material for producing various sodium and chlorine compounds.
This interactivity helps users understand the position of sodium and chloride ions in halite and their ratio in the crystal structure.



Interactive 3. Structure of Fluorite (CaF_2).

More information for interactive 3

An interactive three-dimensional model depicts a ball-and-stick model of fluorite (CaF_2). This model features a cubic structure where calcium ions occupy the corners and the centers of each face of the cube and fluoride ions occupy the tetrahedral voids within the cubic lattice. Each fluoride ion is connected to four calcium ions.
Users can rotate the model to view it from different angles and zoom in or out using the mouse scroll. In the model, the calcium and fluoride ions are displayed as smaller and larger spheres, respectively.
A panel on the left provides view options, including the entire model, fluor, and calcium. Selecting fluor or calcium highlights their arrangement within the model.
On the right side, options for changing the background color and option for zooming the interactive are given.
At the bottom, options for introduction, share, screenshot, augmented reality, and language settings are available.
The introduction option displays the following details:
Chemical name: Calcium fluoride
Molecular formula: CaF_2
Molar mass: 78.07 g/mol
Mohs scale hardness: 4
Crystal system: Face-centered cubic lattice of Ca^{2+} ions, with F^- ions occupying the tetrahedral cavities.
Crystal habit: Typically forms cubic crystals, occasionally exhibiting octahedral shapes.
Description: Pure calcium fluoride is a colorless crystalline substance that is insoluble in water. In nature, the presence of trace amounts of rare earth elements gives fluorite a variety of colors, ranging from blue, yellow, and white to purple and even black.
Applications: Fluorite (calcium fluoride) is used in the production of hydrofluoric acid. It is also used in metallurgy and the glass industry. Additionally, because of its optical properties, calcium fluoride is commonly used as a material for manufacturing lenses and other optical components.
This interactivity helps users understand the position of calcium and fluoride ions in fluorite and their ratio in the crystal structure.

In **Interactive 3**, the lattice structure of two minerals, halite and fluorite, are shown. Halite is the ionic compound NaCl. Notice how there is an equal ratio of sodium ions to chloride ions in the lattice structure. Fluorite is the ionic compound CaF_2. Notice how there are two times as many fluoride ions as there are calcium ions in the lattice structure.

International Mindedness

The term lattice is used in science to denote a regular repeating three-dimensional structure of ions, atoms, or molecules, but it is also used in other contexts. Lattice multiplication is a specific technique of multiplying numbers using a grid. This method of multiplication has been used historically by many different cultures. For example, there is evidence of this in Arabic, Chinese and European mathematics.

Strength of the ionic bond

The strength of an ionic bond varies between different compounds. Lattice enthalpy is a measure of the strength of the electrostatic attraction between ions in an ionic compound. You will learn more about lattice enthalpy in section R1.2.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/born-haber-cycle-hl-id-43579/). Values of lattice enthalpy can be found in section 16 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lattice-enthalpies-at-29815-k-experimental-values-id-45118/) of the DP chemistry data booklet.

- The larger the lattice enthalpy the stronger the ionic bond.
- The lower the lattice enthalpy the weaker the ionic bond.

Table 1 shows a summary of the lattice enthalpies for group 1 and 2 chlorides. Looking at this data, can you think of any factors that would influence the strength of the ionic bond?

Table 1. Lattice enthalpies and cationic radius of group 1 and

Group 1 chlorides	Radius of cation (10^{-12} m)	Lattice enthalpy (kJ mol^{-1})	Group 2 chlorides	Radiu cation (m)
LiCl	76	864	$BeCl_2$	45
NaCl	102	790	$MgCl_2$	72
KCl	138	720	$CaCl_2$	10(
RbCl	152	695	$SrCl_2$	118
CsCl	167	670	$BaCl_2$	135

The box below mentions a different, but similar concept to lattice enthalpy that you will encounter later on in the course.

Making connections

In section R1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-breaking-and-forming-id-43473/) you learn about bond enthalpy. Bond enthalpy is a similar concept to lattice enthalpy but a measure of the strength of covalent bonds. Both are quantitative, energetic measurements. You learn more about quantitative, energetic measurements known as enthalpy in section R1.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/).

Radius of the ions

Recall from section S1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) that the atom is mostly empty space. The radius of an atom or ion is determined dependent on the number of occupied energy levels. The more occupied energy levels there are, the larger the radius of an ion. Do you think larger or smaller ions would have a stronger attractive force with one another?

Looking at **Table 1**, you can see, for group 1 chlorides, as the radius of the cation increases, the lattice enthalpy decreases. This means that the larger the radius of the ion, the weaker the ionic bond. This is because of the greater distance between the two ions as the radius increases, weakening the electrostatic attraction.

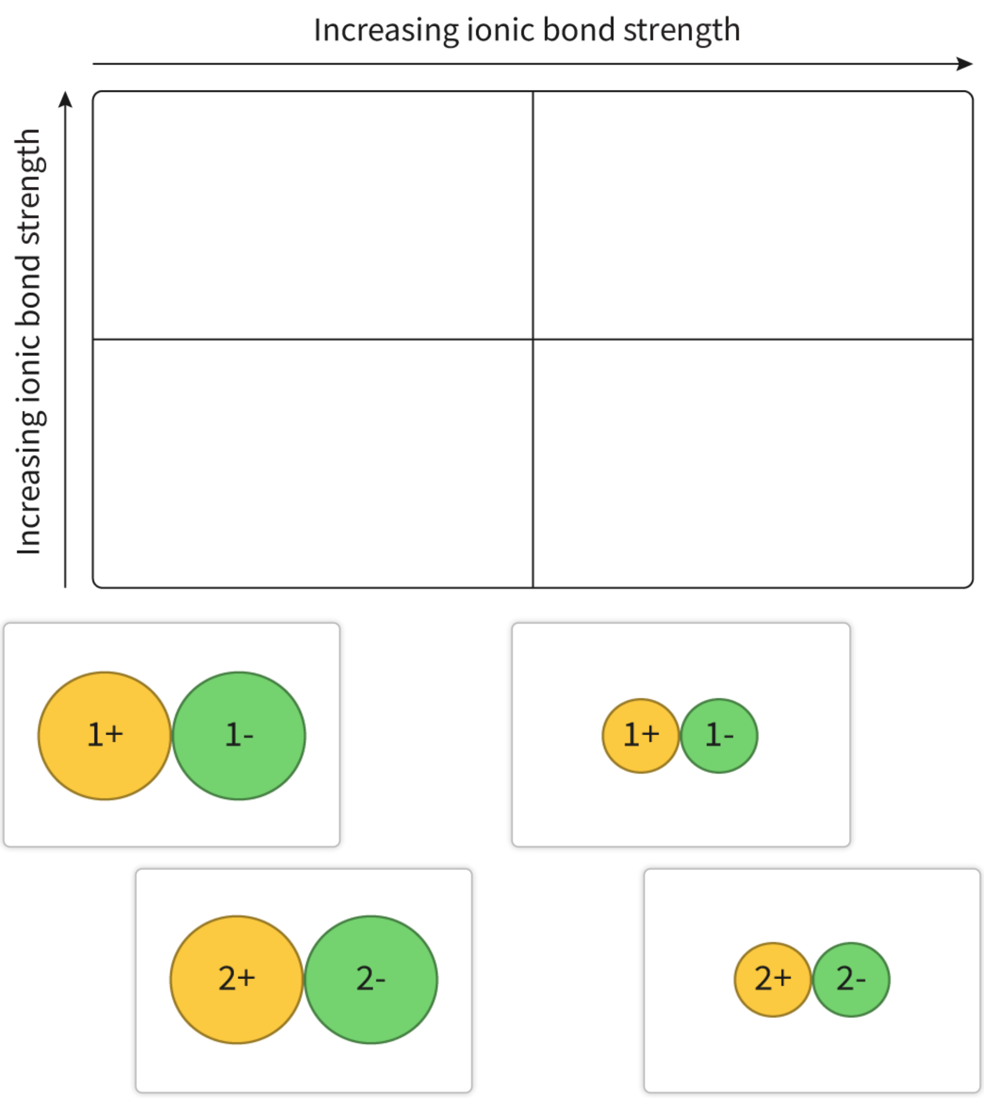
Charge of the ions

Recall that the ionic bond is the electrostatic attraction between oppositely charged ions. How do you think the charge on cations and anions would affect the attractive forces between them?

Looking at **Table 1**, you can see that group 2 chlorides have considerably higher lattice enthalpies than group 1 chlorides. This is due to the greater charge on the cation. Group 2 cations would have a 2+ charge whereas group 1 ions have a 1+ charge. This greater charge results in a stronger electrostatic attraction and a stronger ionic bond.

In summary, the smaller the radius and the greater the charge of the ions, the stronger the ionic bond. The larger the radius and the lower the charge of the ions, the weaker the ionic bond.

In **Interactive 4,** drag and drop pairs of ions according to their ionic bond strength.



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Interactive 4. Pairs of Ions According to Their Ionic Bond Strength.

More information for interactive 4

An interactive 2×2 grid challenges users to analyze and arrange different ion pairs based on their ionic bond strength. The grid is guided by directional arrows: ionic bond strength increases from left to right (→) and from bottom to top (↑), encouraging learners to make comparisons based on both ion size and ionic charge.
Users must drag and drop four distinct ion pair models into the appropriate grid cells.

Kog

The models provided are as follows:
A large yellow sphere with a 1+ charge is in contact with a large green sphere with a 1- charge.
A large yellow sphere with a 2+ charge is in contact with a large green sphere with a 2- charge.
A small yellow sphere with a 1+ charge is in contact with a small green sphere with a 1- charge.
A small yellow sphere with a 2+ charge is in contact with a small green sphere with a 2- charge.
This activity visually reinforces the concept that ionic bond strength is influenced by ionic size and charge, with stronger bonds forming between smaller ions and ions with higher charges.
Read below for the solution:
Column 1, Row 1: A small yellow sphere with a 1+ charge is in contact with a small green sphere with a 1- charge.
Column 2, Row 1: A small yellow sphere with a 2+ charge is in contact with a small green sphere with a 2- charge.
Column 1, Row 2: A large yellow sphere with a 1+ charge is in contact with a large green sphere with a 1- charge
Column 2, Row 2: A large yellow sphere with a 2+ charge is in contact with a large green sphere with a 2- charge.

The box below links the two factors that influence the strength of the ionic bond to electrostatic forces, a concept studied in physics.

Making connections

The electrostatic interaction between two charged particles is an electrostatic **force**, a concept studied in physics. Coulomb's law mathematically describes the electrostatic force between two charged particles:

$$F = k\frac{q_1 q_2}{r^2}$$

Where F represents the electrostatic force, k is Coulomb's constant, q_1 and q_2 are the charges of each of the charged particles and r is the distance between the two charged particles. By looking at this equation, it is easy to see that a greater charge and shorter distance will result in a greater force. These were the two factors considered before for the strength of the ionic bond.

For this activity you will investigate the structure of ionic compound using models.

Activity

- **IB Learner profile attribute:** Thinkers
- **Approaches to Learning:**
    - Thinking — Designing models
    - Research — Evaluating information sources
- **Time required to complete activity**: 30 minutes
- **Activity type:** Pairs, followed by whole class discussion

Make an ionic model!

What you need:

- Plasticine/modelling clay
- Straws/toothpicks

Then follow these instructions:

1. In pairs, create model representing the ionic lattice of:
   (a) Sodium chloride
   (b) Magnesium chloride
2. Discuss with your partner which ionic compound you would expect to have stronger ionic bonds and why.
3. After the creation of your models, research the structure for each of these.
4. As a class, consider the following questions.
   (a) What parts of your structure were similar to the researched structure?
   (b) What parts of your structure were different from the researched structure?
   (c) How would you assess the credibility of the source you obtained information from?
   (d) What part of this task did you find simple?
   (e) What part of this task did you find difficult? Why?

S2.1 The ionic model

Physical properties of ionic compounds

Learning outcomes

By the end of this section you should be able to explain the physical properties of ionic compounds in connection to its structure and bonding for the following:

- melting point
- volatility
- solubility in water
- electrical conductivity
- brittleness.

Ionic compounds are commonly found as salts, minerals and ceramics. Society typically uses the term salt to refer to sodium chloride, however, in chemistry this term is used to specify an ionic compound. While you might be most familiar with sodium chloride (table salt), salts exist in many different colours depending on the nature of the cation, anion or even the solvent within the lattice structure. Scroll through the images in **Interactive 1** to see some different examples of salts you may encounter in the course.

Interactive 1. Examples of Different Salts.

More information for interactive 1

Thinking about table salt (NaCl), an ionic compound you are likely familiar with from everyday life, what are some properties you know about this salt? If you add it to a pot of water, will it dissolve? Does it melt easily? Does it break easily into small pieces? Can it conduct electricity in its solid state? What about as a solution in water? Think about your experiences with table salt as we look more closely at the physical properties of ionic compounds.

Melting points

Think about all the examples of ionic compounds we have seen so far. What state of matter do you notice most of them have been in? Nearly every example of ionic compound we have looked at has been in its solid state. In section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) you learned what change of state occurs during the process of melting. A melting point is the temperature at which a solid will undergo a change of state to a liquid. Would you expect substances that are solid at room temperature to have high or low melting points?

Melting an ionic compound to form a liquid involves breaking the electrostatic attractive forces between the oppositely charged ions in the lattice. Due to the strength and number of the electrostatic attractive forces between ions, a lot of energy is required to melt ionic compounds. Therefore, ionic compounds have high melting points. **Table 1** summarises some of the melting points for group 1 and 2 chlorides. Notice that these values are all quite high, much higher than can be achieved with an oven in your home!

Table 1. Melting points for group 1 and 2 chlorides.

Group 1 chlorides	Melting point (°C)	Group 2 chlorides	Melting point
LiCl	605	$BeCl_2$	415
NaCl	801	$MgCl_2$	714
KCl	770	$CaCl_2$	772
RbCl	718	$SrCl_2$	874
CsCl	645	$BaCl_2$	962

The box below looks at a group of ionic compounds that have lower melting points than those typically referenced when discussing ionic compounds.

International Mindedness

Ionic liquids are a group of ionic compounds with significantly lower melting points (<100 °C). Ionic liquids are unique in that they can dissolve a large range of different solutes. Globally, many current solvents used in industry are flammable and give off volatile compounds that are harmful to the surrounding environment. Using ionic liquids as solvents reduces these concerns. Ionic liquids form an active area of research in Green Chemistry, an area of chemistry focused on the minimisation and elimination of hazardous substances. Visit this source (https://www.electrochem.org/dl/interface/spr/spr07/spr07_p38.pdf) if you are interested in learning more about ionic liquids.

Volatility

The term volatility describes the ease at which a substance vaporises or becomes a gas. In section S1.1.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/) you learned which changes of state are involved in volatility. Volatility would be the ease at which a substance evaporates, boils or sublimates. A substance with a high volatility easily becomes a gas. You have likely encountered volatile substances as the 'smelly' substances in vinegar, perfume and paints. A substance with a low volatility does not become a gas easily. As ionic compounds have high melting points and do not easily vaporise they have a low volatility.

There are, however, substances known as smelling salts. These salts are usually made of ammonium carbonate and were traditionally used to relieve faintness due to the odour given off. Recall what you learned about the strength of ionic bonds and lattice enthalpy in section S2.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/). How do you think the lattice enthalpy of smelling salts compares to that of typical salts?

Solubility

Have you ever noticed that when you shake table salt into water it dissolves? From section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/), what is the name of the homogeneous mixture formed when table salt dissolves in water? When sodium chloride in table salt dissolves in water an aqueous solution is formed. Sodium chloride dissolves in water due to the strong interactions sodium and chloride ions have with water compared to between the oppositely charged ions in the lattice. These strong interactions allow water to surround individual ions, separating ions from their fixed positions in the lattice. **Figure 1** shows how a solution is formed from solid sodium chloride when it encounters water.

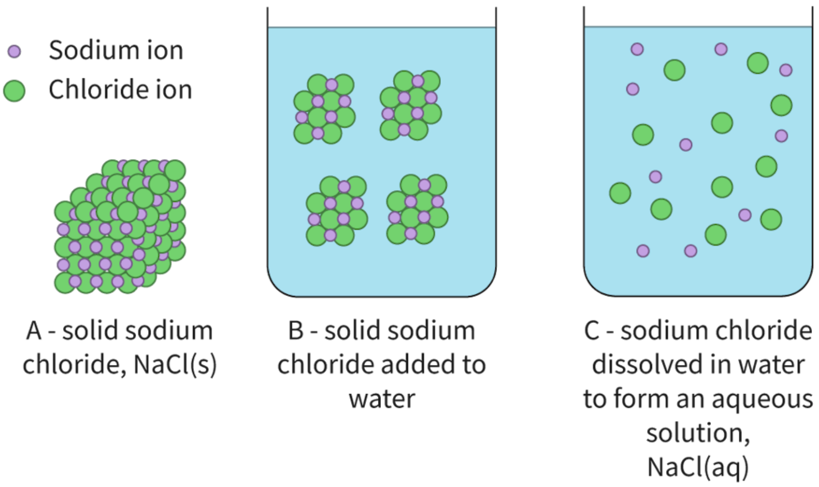


Figure 1. Solid sodium chloride dissolving in water.

More information for figure 1

We can represent the physical change occurring during the formation of an aqueous solution for sodium chloride using state symbols as:

$$NaCl(s) \rightarrow NaCl(aq)$$

Since individual ions are surrounded by water in the aqueous solution of an ionic compound, we can also write the physical change occurring as:

$$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

Many, but not all, ionic compounds are soluble in water. The solubility of an ionic compound depends on the strength of the attraction between the ions and water along with the strength of the ionic bond. An ionic compound is insoluble in water when it does not form a homogeneous mixture when added to water.

If you try to shake table salt onto cooking oil, you might notice it does not dissolve. This is because there are no strong interactions between cooking oil and sodium and chloride ions. This makes sodium chloride insoluble in oil. What type of mixture is formed when an ionic compound is insoluble in a solvent?

Theory of Knowledge

Nacre, also known as mother of pearl, is produced on the inner shell of some oysters when a piece of foreign organic material, such as a parasite, enters the shell, or if the oyster's tissue is damaged by a predator. Nacre is composed of layers of calcium carbonate held together by organic material, producing a strong and beautifully iridescent substance highly valued in the art and jewellery industries. Historically, pearl oysters were collected by hand, opened and inspected for pearls — a very tedious practice as finding pearls is very rare and also wasteful, greatly depleting the pearl oyster population. Today, 95% of pearls available are 'cultured' by placing a small bead inside an oyster's shell and allowing the oyster to cover the bead with layers of nacre. The resulting pearl is produced faster and with a more consistent shape. This process, developed in 1956, also allowed the pearl oyster population to regenerate to a level that they no longer face extinction. Cultured pearls and natural pearls are almost identical in appearance — the only way to be absolutely certain is to examine the pearl by X-ray. Does the ability to replicate a natural process, producing a nearly identical product, reduce the beauty or value of nacre?

Conductivity

The term electrical conductivity refers to the ability of charged particles to move through a region of space. Recall what you know about ionic compounds. What charged particles are involved? Are these charged particles able to move? Can you think of situations where these charged particles can move?

Solid ionic compounds have cations and anions as charged particles that are fixed in the crystal lattice. This lattice structure in the solid state prevents movement of the charged ions. Therefore, solid ionic compounds are not electrically conductive.

When an ionic compound melts to form a molten liquid, the ionic bonds holding oppositely charged ions begin to break. This allows cations and anions to move throughout the liquid. Since there is motion of charged particles, molten ionic compounds are electrically conductive.

When an ionic compound dissolves in water to form an aqueous solution, ions are surrounded by water. Since ions are no longer fixed in positions (as they were in the solid lattice) they are free to move throughout the solution. As there is motion of charged particles, aqueous ionic compounds are electrically conductive. **Figure 2** illustrates the conductivity for ionic compounds in various states using an electrical circuit.

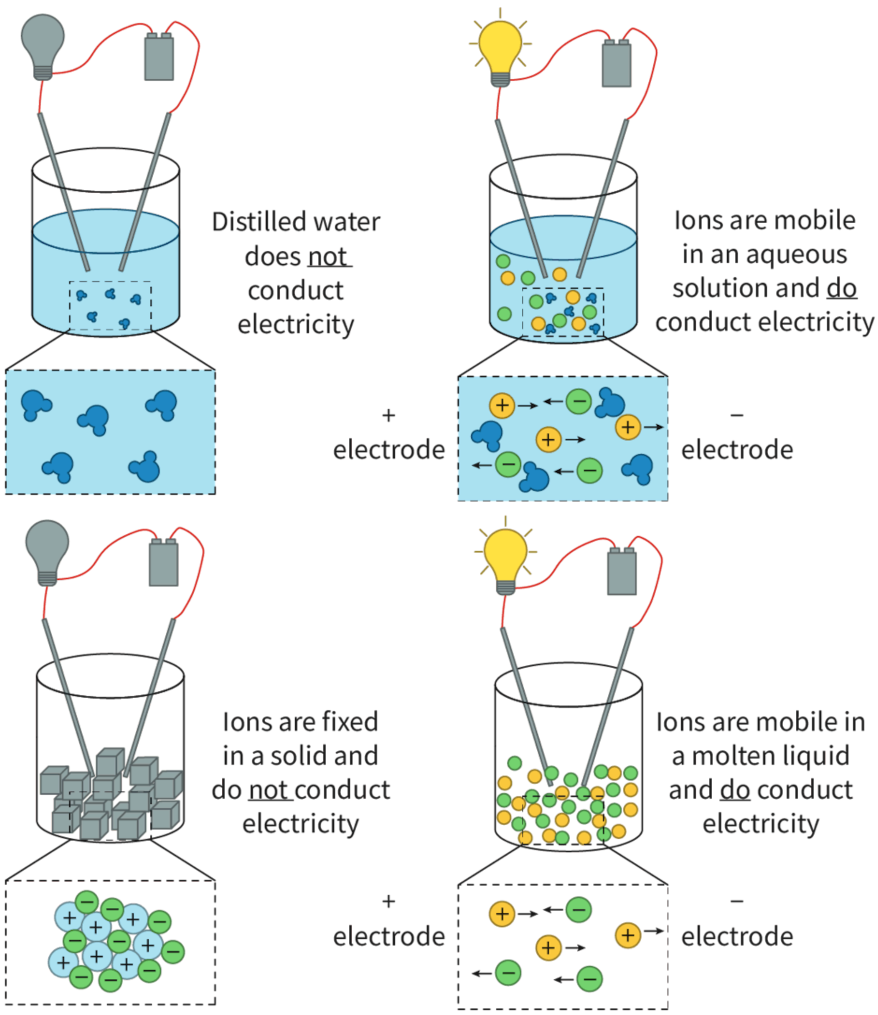
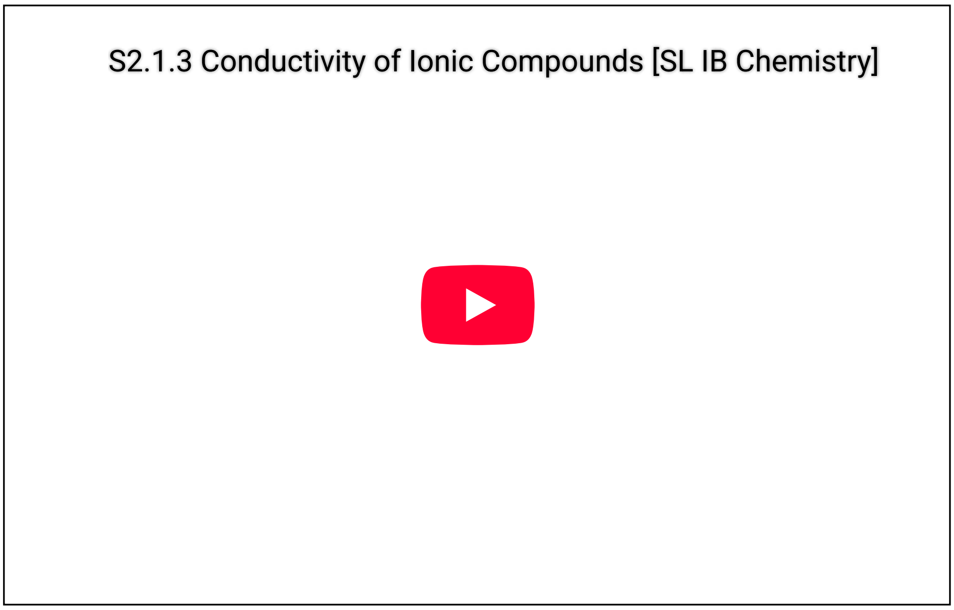


Figure 2. Electrical circuit showing when ionic compounds are electrically conductive.

More information for figure 2

Watch **Video 1** to help you visualise the conductivity of ionic compounds in the states of matter listed above.



Video 1. Visualise the conductivity of ionic compounds in the solid, molten and aqueous states.

The box below highlights common misconceptions of students when it comes to the conductivity of ionic compounds.

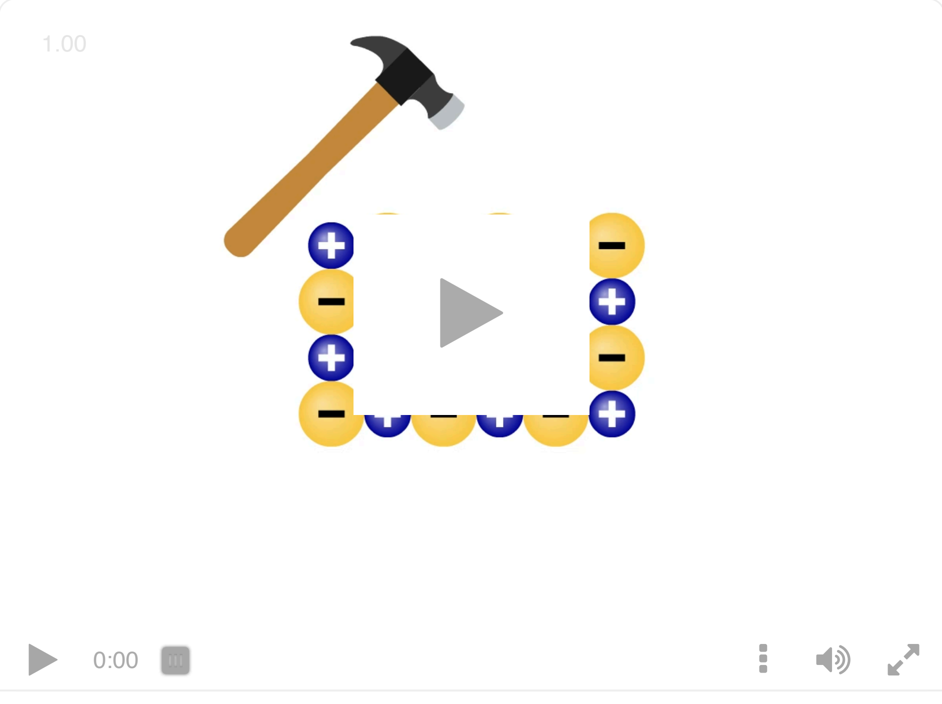
Study skills

Conductivity refers to the mobility of charged particles. It is very common for students to misinterpret conductivity to be exclusively due to electrons, however, in the case of aqueous ionic compounds, the conductivity of the solutions is from the mobility of the ions in solutions, **not** electrons.

Brittleness

Have you ever noticed if you have a large piece of table salt it can easily be ground into smaller pieces? This happens because sodium chloride is brittle. This is the final characteristic property of ionic compounds. While ionic bonds are strong, ionic substances fracture easily. These properties can be explained by taking a close look at the lattice structure as shown in **Interactive 2**.

When a stress is applied (caused by dropping or hitting with a hard object), layers within the lattice shift in response to that stress. As layers shift, ions become adjacent to like charges. These like charges experience an electrostatic repulsion and layers separate from one another, which forces the lattice to break into pieces. This is why ionic compounds, like sodium chloride in table salt, are brittle.



Interactive 2. Ionic Lattice Shifting in Response to an Applied Stress.

More information for interactive 2

The box below provides some examples of methods to collect qualitative and quantitative data for the properties of ionic compounds discussed in this section.

Practical skills

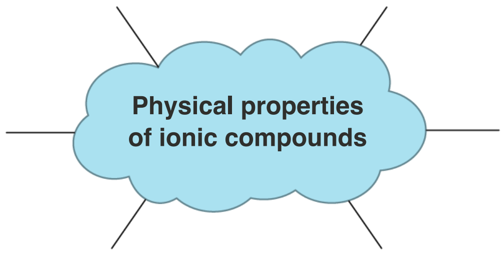
Inquiry 2: Collecting and processing data — Collecting data

There are qualitative and quantitative methods of collecting data to study the physical properties of ionic compounds. Below are examples of a few ways you could collect qualitative and quantitative data to study some of these physical properties.

	Qualitative data	Quantitative data
Meltin g point	Observe whether or not the substance melts when placed into the flame of a Bunsen burner.	Observe what temperature the substa melts at using a melting point appara
Solubi lity in water	Observe whether a small amount of the solute forms a homogeneous mixture when added to water.	Use a conductivity meter to measure h conductivity of the solution compares t water. If the ionic compound is colour spectrophotometer can also be usec measure the solubility.

	Qualitative data	Quantitative data
Electrical conductivity	Observe whether a lightbulb of a simple electrical circuit lights up.	Use a conductivity meter to measu conductivity of solution.
Brittleness	Observe (macroscopically with eyes or microscopically with microscope) the ease at which cracks and fractures form on the crystal's surface.	Use a structures and materials teste measures the force at which the mat fractures.

Drag and drop the boxes in **Interactive 3** to see whether you can recall all the physical properties of ionic compounds.



conducts electricity in solid state	low volatility	low melting points	none are soluble in water
conducts electricity in molten state	high volatility	high melting points	some are soluble in water
conducts electricity in aqueous state	brittle	flexible	all are soluble in water

Check

Interactive 3. Summary of the Physical Properties of Ionic Compounds.

More information for interactive 3

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

According to the USGS (https://www.usgs.gov/faqs/why-ocean-salty#:~:text=The%20concentration%20of%20salt%20in,be%20about%20120%20million%20tons.), there are about 120 000 000 tons of salt in a cubic mile of seawater (which is equivalent to 1.09×10^{11} kg in 4.2 km^3 of seawater) and more than 90% of this salt is sodium chloride, an ionic compound. The world's oceans are a vast source of biodiversity but they are at risk from a variety of threats such as plastic pollution and overfishing.

8 June is World Oceans Day (https://www.un.org/en/observances/oceans-day) which aims to increase the public's perception of the threats posed to the world's oceans. What can you do at your school or in your local community to raise awareness of World Oceans Day?

In the following activity you will check your understanding of the physical properties of the ionic compounds covered in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning**; Thinking skills — Combining different ideas in order to create new understanding
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity for steps 1—5, pairs for step 6

1. Looking at the chemical formula for the substances below. Identify which ones you suspect are ionic. Explain how you made your selection.
(a) $MgCl_2$
(b) CO_2
(c) NH_4Br
(d) Na
2. Research the melting point of each of the substances above. Comment on how you might use the melting point to confirm or refute your selection above.
3. Explain why ionic compounds are brittle using a diagram to support your answer.

4. Sketch a diagram showing what you think happens at the atomic scale when sodium chloride dissolves in water.
5. Create a Venn diagram comparing solid ionic compounds to aqueous ionic compounds. Include structure, properties and diagrams.
6. Compare your work with a peer. Highlight three areas your work had in common. Highlight three areas where your work differed. Discuss these differences with your peer.